

An Electromechanical Display with Tactile Feedback Using Multimodal CNN–BiLSTM Learning

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Abstract—Assistive technologies face major difficulties when they need to enable non-visual communication in settings where people cannot use visual displays or audio signals during situations with low visibility and high noise levels and hands-free requirements. The current tactile interfaces only deliver basic vibration notifications which lack ability to show different meanings while most multimodal assistive systems provide their main output through visual and auditory channels which need high-performance computational systems and expensive equipment. The current study demonstrates limitations by showing how assistive tools can only be used in budget-friendly situations which require their operation to happen almost immediately. The study proposes a new multimodal learning method which transforms visual and textual input into meaningful tactile output through an electromechanical system that uses vibrations to create tactile feedback. A lightweight convolutional neural network encodes visual information while BERT-based embeddings enable textual input processing followed by BiLSTM which extracts both contextual and sequential data. The fused multimodal representation is classified into three semantic categories—Instruction, Alert, and Information—which are mapped to distinct vibration patterns created through Arduino-controlled vibration motors.

Experimental evaluation demonstrates that the proposed approach achieves a test classification accuracy of 91%, with an overall precision of 0.90, recall of 0.91, and F1-score of 0.90. User-level tactile interpretation experiments further indicate that 92% of the vibration patterns were correctly identified after brief familiarization, confirming the clarity and usability of the tactile mappings. These results show that integrating lightweight multimodal learning with affordable electromechanical hardware enables effective and interpretable non-visual communication suitable for near real-time assistive applications.

Keywords—Multimodal Learning, Tactile Feedback, Assistive Interfaces, Convolutional Neural Network, BiLSTM, Vibration-Based Haptic Interface

I. INTRODUCTION

The latest developments in assistive technology research aim to create better access solutions for users who work in environments with visual challenges and high noise levels and require hands-free operation and for users who have visual disabilities. Human-computer interaction methods today depend on visual displays and audio feedback which only work

when the user environment remains normal [2], [3]. Users in noisy settings cannot hear audio cues and users in situations with limited visibility and critical visual requirements find visual interfaces extremely challenging to operate.

The current assistive technologies provide users with visual and audio output options which include screen readers and camera-based scene interpretation and speech-driven alerts [6], [7]. The methods deliver extensive contextual insights but require users to maintain constant visual focus or full audio access which does not reflect actual conditions in most environments. The need for non-visual communication methods has become more prominent.

Tactile interfaces provide a promising non-visual communication method which researchers have tested through vibration feedback systems that deliver alerts and notifications [3]. The current tactile systems use fixed vibration patterns for their operation because they do not have the ability to understand different types of information, which prevents them from delivering specific instructions and warning messages and general information.

The models from multimodal learning systems which use both visual content and textual data show better semantic understanding results than models which rely on a single mode of input [4], [5]. The models achieve better classification results because they utilize both visual content and textual information to eliminate uncertainty from their predictions. Multimodal systems produce their main results through either visual or audible content whereas they treat tactile elements as secondary means of communication.

There exists a research gap which needs to be filled through the combination of multimodal semantic understanding with vibration-based tactile feedback systems that allow people to communicate without using visual cues. The existing research gap needs a solution which can simultaneously handle image-text data while converting semantic meanings into organized tactile feedback through budget-friendly hardware and lightweight models.

The recent studies show that multimodal learning systems which combine visual and textual content through their

TABLE I
COMPARISON OF RECENT ASSISTIVE AND TACTILE COMMUNICATION APPROACHES

Work	Input Modality / Dataset	Algorithms Used	Output / Hardware	Real-Time Suitability
Bettelani <i>et al.</i> [1]	Text (Braille symbols)	Rule-based control, solenoid actuation	Electromagnetic Braille display	Limited (mechanical refresh delay)
Chen <i>et al.</i> [2]	Control signals	Pattern-based actuation logic	Electromagnetic coil-based tactile display	Yes (fast actuation)
Alvarez <i>et al.</i> [3]	Control commands	Micro-actuator control logic	Vibration-based tactile interface	Yes (low-latency actuation)
Singh <i>et al.</i> [4]	Image-Text pairs (custom dataset)	CNN + Transformer-based text encoder	Visual / symbolic output	Moderate (model complexity dependent)
Huang <i>et al.</i> [5]	Image-Text pairs	Lightweight CNN + attention-based fusion	Visual / auditory output	Moderate (fusion overhead)
Morales <i>et al.</i> [6]	Text and multimodal inputs	Transformer-based language models	Audio / textual guidance	Limited (high inference cost)
Yang <i>et al.</i> [7]	Multimodal inputs	Vision-language model with multimodal reasoning	Visual / auditory assistive output	Limited (computational overhead)
This work	Image-Text pairs (Flickr8k)	Lightweight CNN + BERT + BiLSTM	Arduino-controlled vibration motors	Suitable (lightweight inference and fast actuation)

learning frameworks produce successful results for assistive and accessibility applications because the systems enhance both contextual understanding and interaction reliability [4]–[6], [10], [13], [18], [20].

II. RELATED WORK

The field of research on non-visual assistive communication investigates three main areas which include tactile interface development and vibration feedback systems and multimodal educational systems. The initial research on tactile communication established its foundation through hardware development which centered on refreshable Braille displays and electromagnetic actuation systems. The research by Bettelani *et al.* [1] describes an electromagnetic Braille display system which uses solenoid actuation to produce tactile characters, while the study focuses on mechanical reliability and readability instead of evaluating the system’s ability to understand input data.

Researchers have studied electromagnetic and vibration-based tactile interfaces to replace traditional visual and auditory methods of communication. The research by Chen *et al.* demonstrated that electromagnetic tactile displays can produce consistent tactile patterns through coil-based actuation, while their system operates with physical feedback that depends on established patterns which lack semantic meaning. Alvarez *et al.* created vibration-based tactile systems which use micro-actuators to provide non-visual notifications, their system achieves rapid response capability but delivers only basic status updates to users.

Multimodal learning systems have developed new methods for semantic comprehension through their ability to process visual and textual information simultaneously. The multimodal framework proposed by Singh *et al.* combines convolutional neural networks with transformer-based language models to achieve better accuracy when classifying image-text datasets. Huang *et al.* used lightweight CNN architectures and attention-based mechanisms to study vision-language fusion, they found that their method improved contextual reasoning

compared to unimodal approaches. The systems produce outputs which consist of visual and symbolic content that serves different purposes beyond their current functions.

Large language model-based assistive systems have also been investigated for accessibility applications. Morales *et al.* presented an assistive interaction framework leveraging transformer-based language models to generate structured guidance for users, while Yang *et al.* introduced a multimodal assistive system that integrates visual and textual reasoning for accessibility support. Although these approaches demonstrate strong semantic reasoning capabilities, they depend on computationally intensive models and predominantly generate visual or auditory outputs.

In contrast to the above works, the approach presented in this paper focuses on lightweight multimodal semantic inference with vibration-based tactile feedback as the primary output modality. These limitations motivate the approach presented in this work.

Vibration-based and electromagnetic tactile interfaces have been extensively investigated for non-visual communication and assistive interaction, including wearable haptic systems, tactile displays, and vibrotactile feedback mechanisms designed for accessibility and human-computer interaction [1]–[3], [9], [11], [14], [17], [19].

III. PROPOSED SYSTEM

This section describes the dataset, multimodal feature extraction pipeline, fusion strategy, hardware components, and testing setup used to generate vibration-based tactile feedback from image-text inputs.

A. Dataset Description

The multimodal learning model was trained and evaluated using the Flickr8k dataset, a publicly available image-text corpus commonly used for vision-language research. The dataset consists of 8,092 images collected from Flickr, with each image paired with five human-annotated textual captions describing the visual content. The total collection

contains around 40,000 image-caption pairs. The research establishes separate multimodal samples which include every image-caption pair as distinct entities. The dataset was divided into three parts which included training and validation and testing according to a 70%–15%–15% ratio. The division resulted in the acquisition of 5,664 training images together with 1,214 validation images and 1,214 testing images. The team kept all image captions during the dataset division process because they needed to prevent data leakage.

The process of feature extraction began after all images underwent resizing to 224×224 pixels and normalization to the $[0, 1]$ range. The textual captions were tokenized using a BERT tokenizer. The average caption length after tokenization was approximately 10–12 tokens with padding applied to ensure uniform sequence length across samples.

B. Feature Extraction and Processing Pipeline

The multimodal pipeline design begins with separate processing for visual and textual inputs which later combines these two types of data into a single representation. The system maintains separate processing for each modality to protect its unique features during the process of extracting different features. The proposed multimodal fusion pipeline produces output predictions for three semantic categories which include Instruction, Alert, and Information. The fusion model uses image and caption data to examine visual content and textual elements which it uses to predict one of three semantic intent classes based on its learned multimodal knowledge. The dataset does not include these classes as predefined annotations because they were created through weakly supervised labeling which was done before training for the purpose of generating downstream tactile feedback. The visual encoder uses a custom lightweight convolutional neural network that includes convolutional and pooling layers which operate to extract features but not to perform deep classification tasks.

1) *Visual Feature Extraction:* The system extracts visual elements through a specially designed light-weight convolutional neural network which operates with minimal processing requirements. The network comprises multiple convolutional layers which use ReLU activation together with max-pooling operations, followed by a flattening layer that generates a constant-length feature vector. The CNN serves as a feature extractor which does not conduct independent classification during multimodal training.

2) *Textual Feature Extraction:* The system uses pre-trained BERT embeddings to obtain textual features which provide contextual word representations. The system processes embedding sequences through a Bidirectional Long Short-Term Memory (BiLSTM) network to establish both sequential dependencies and contextual relationships present in the caption. The BiLSTM system uses its final hidden states to create the textual feature representation.

C. Multimodal Fusion Strategy

The system combines visual and textual features through a process that uses feature-level concatenation as its unifying

method. The visual feature vector v exists in the space of real numbers while the textual feature vector t exists in the space of real numbers according to their respective dimensionalities which are d_v and d_t . The multimodal fusion process generates the combined representation f through the following equation:

$$f = [v \parallel t]$$

The operation of vector concatenation gets represented through the symbol $\parallel$. A multilayer perceptron (MLP) system processes a combined feature vector which performs both dimensionality reduction and classification tasks. The softmax layer generates class probabilities which correspond to three semantic categories that include Instruction Alert and Information.

This fusion method enables both modalities to supply their distinct information for the ultimate prediction while maintaining computational requirements at a level which enables near real-time evaluation. The system used feature-level fusion because this method provides better semantic representation while requiring less processing power than late-stage decision fusion and attention-based fusion methods. The model processes both visual and textual input together before making classification decisions which lets both methods contribute to understanding semantic intent but does not require resource-intensive cross-modal attention methods. The design makes products which support lightweight systems that need to provide assistive functions through their capacity to work quickly and implement functions without difficulty.

D. Semantic Label Definitions

The proposed system classifies multimodal inputs into three semantic intent categories based on the contextual meaning conveyed by the combined image and text information. *Instruction* refers to inputs that imply guidance or suggested actions for the user. *Alert* corresponds to inputs indicating potentially critical, cautionary, or attention-demanding situations. *Information* includes descriptive inputs that convey general scene content or contextual details without implying urgency or required action. These definitions are used consistently throughout model training, evaluation, and tactile feedback mapping.

In the proposed architecture, the convolutional neural network functions exclusively as a visual feature extractor and does not perform standalone classification. The system generates a compact representation of an input image by encoding its spatial and semantic information. BERT embeddings serve as the foundation for textual processing because they enable the extraction of contextual word-level meanings which BiLSTM models use to analyze sequential dependencies and sentence structures in the caption. The system uses pretrained language representations for its first function, while it maintains its ability to model time-based data through its recurrent layer functionality. The system achieves multimodal integration by combining visual and textual feature vectors at the feature level, which prevents the creation of unnecessary model complexity.

TABLE II
ILLUSTRATIVE EXAMPLES OF SEMANTIC LABEL GENERATION FROM
IMAGE CAPTIONS

Caption Attribute	Example Caption	Assigned Label	Labeling Rationale
Action-oriented instruction	"A person crossing the road should look both ways before moving forward."	Instruction	Presence of action verbs implying guidance or recommended behavior
Cautionary or hazardous context	"A cyclist rides close to moving traffic on a busy street."	Alert	Caption indicates a potentially risky or attention-demanding situation
Purely descriptive scene	"A dog running freely in a grassy park under clear skies."	Information	Descriptive content without urgency or implied action

E. Semantic Label Generation

Since the Flickr8k dataset does not provide predefined semantic intent labels, a weakly supervised labeling strategy was employed to derive the Instruction, Alert, and Information classes. Captions were assigned semantic labels using a rule-based mapping based on linguistic cues and contextual indicators present in the text. Captions implying guidance or actions were labeled as Instruction, captions indicating cautionary or attention-demanding situations were labeled as Alert, and purely descriptive captions were labeled as Information. These derived labels were used as reference annotations for model training and evaluation.

Table II provides illustrative examples of how semantic intent labels are derived from caption attributes using the proposed weakly supervised labeling strategy.

F. Model Training and Testing Setup

The multimodal classification model was trained using the Adam optimizer with categorical cross-entropy loss. Training was conducted for a fixed number of epochs with early stopping applied based on validation loss to prevent overfitting.

Model evaluation was conducted exclusively on the held-out test set. Performance was measured using accuracy, precision, recall, and F1-score. In addition, a confusion matrix was computed to analyze class-wise performance and misclassifications patterns. A batch size of 32 and an initial learning rate of $1e4$ were used during training, with early stopping applied based on validation loss.

G. Hardware Components and Implementation Details

The tactile feedback module consists of an Arduino Uno microcontroller interfaced with vibration motors through NPN transistors and protective circuitry. The vibration motors used are coin-type ERM (Eccentric Rotating Mass) motors operating at 5 V. NPN transistors (e.g., BC547) are employed as switching elements to safely drive the motors, while flyback diodes are used to suppress inductive voltage spikes. Current-limiting resistors are included to ensure safe operation.

Each semantic category is mapped to a distinct vibration duration and pattern to ensure perceptual separability.

TABLE III
SUMMARY OF SOFTWARE AND HARDWARE COMPONENTS

Component	Specification
Dataset	Flickr8k (8,092 images, 5 captions per image)
Image Resolution	224 × 224 pixels
Text Length	Average 10–12 tokens per caption
Visual Model	Custom lightweight CNN (feature extractor)
Text Model	BERT embeddings followed by BiLSTM
Fusion Method	Feature-level concatenation + MLP classifier
Controller	Arduino Uno
Actuator	5 V ERM coin-type vibration motors
Switching Circuit	NPN transistor (e.g., BC547) with flyback diode
Training Platform	Google Colab (CPU/GPU-supported environment)
Testing Method	Held-out test set evaluation

Model training and inference were performed on a system equipped with an Intel-based CPU and NVIDIA GPU support using the Google Colab environment. The tactile feedback execution runs on the Arduino hardware, enabling a clear separation between computation and actuation.

Microcontroller-driven tactile feedback systems are commonly adopted in assistive prototypes due to their low power consumption, flexibility, and ease of integration with embedded sensing and actuation components [15].

H. System Summary

To provide a consolidated view of the proposed approach, Table III summarizes the key software and hardware components used in the system. The multimodal pipeline integrates visual and textual processing modules with a lightweight fusion strategy, followed by an electromechanical tactile feedback interface. The design emphasizes modularity, allowing individual components such as the visual encoder, text encoder, and tactile actuator to operate independently while contributing to the overall semantic inference process.

The use of lightweight neural models and low-cost hardware components ensures that the system remains computationally efficient and suitable for near real-time deployment.

Prior work has emphasized the importance of lightweight model design and energy-efficient processing for time-sensitive assistive systems deployed on resource-constrained platforms [7], [8], [12], [16].

I. Novelty and Differentiation from Existing Work

The proposed approach differs from existing assistive and tactile communication systems in both functional scope and system design. As illustrated in Table IV, prior tactile interfaces primarily focus on delivering predefined vibration alerts or rely on specialized electromagnetic hardware, limiting their ability to convey contextual semantic information. In contrast, the proposed system treats tactile feedback as

TABLE IV
NOVELTY AND DIFFERENTIATION OF THE PROPOSED APPROACH

Aspect	Existing Approaches	Proposed Approach
Primary Output Modality	Visual or auditory outputs; tactile feedback used only for basic alerts	Tactile feedback used as the primary communication modality
Semantic Understanding	Limited or rule-based; no semantic interpretation for tactile output	Semantic classification using multimodal image-text understanding
Input Modalities	Single modality or loosely coupled multimodal inputs	Joint processing of image and text inputs
Learning Architecture	Computationally intensive CNN-Transformer or unimodal models	Lightweight CNN for vision and BERT-BiLSTM for text
Fusion Strategy	Late-stage decision fusion or attention-heavy mechanisms	Feature-level concatenation enabling efficient multimodal integration
Hardware Dependency	Specialized or hardware-intensive tactile displays	Low-cost electromechanical vibration motors with Arduino control
Near Real-Time Suitability	Limited due to high computational or mechanical overhead	Designed for low-latency inference and fast actuation
Extensibility	Tightly coupled to specific output formats	Modular design allowing easy extension to new modalities or patterns

the primary communication modality driven by multimodal semantic inference.

Unlike computationally intensive multimodal frameworks that prioritize visual or auditory outputs, this work employs a lightweight CNN-BiLSTM architecture with feature-level fusion to achieve efficient semantic classification. The integration of affordable electromechanical vibration motors controlled through a microcontroller further distinguishes the system by enabling low-cost, near real-time tactile communication.

IV. SYSTEM ARCHITECTURE AND IMPLEMENTATION

The proposed system design combines different software processing modes with a vibration-based tactile feedback mechanism to facilitate dependable non-visual communication. Image and text inputs are intended for processing, multimodal classification followed by generating distinct vibration patterns through hardware controlled by Arduino.

A. Overall System Architecture

Fig. 1 shows the total framework of the multimodal tactile communication system. The system takes the user's image and text inputs at the beginning.

Visual data is first subjected to a convolutional neural network of low weight which extracts the spatial characteristics of the image and the processing cost is low. The input text is analyzed with the help of transformer-based

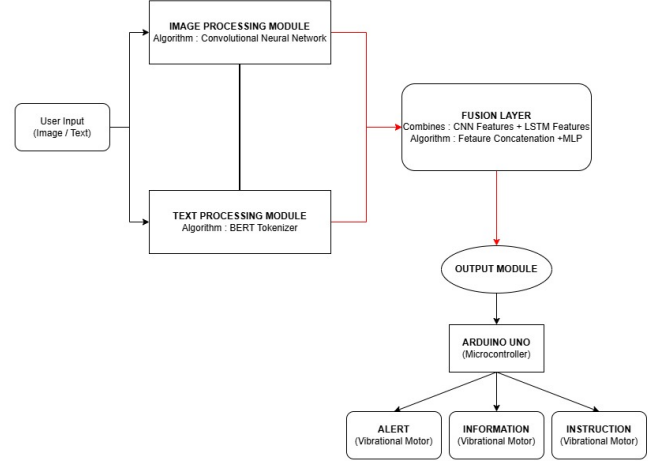


Fig. 1. Overall architecture of the proposed multimodal tactile communication system

language embeddings together with sequence modeling for the purpose of seizing semantic and contextual information. The output that has been predicted is then sent to the tactile feedback module.

B. Hardware Circuit Design

The tactile feedback subsystem consists of an Arduino Uno, vibration motors, NPN transistors, flyback diodes, and current-limiting resistors. The Arduino acts as the main control unit and triggers the vibration patterns corresponding to the classification output received from the software module.

The circuit diagram in Fig. 2 illustrates the safe and effective control of the electromechanical vibration motor. An NPN transistor is used as a switching element, allowing the Arduino to control the motor without exceeding its current-driving capability. A flyback diode is connected across the motor terminals to suppress voltage spikes caused by inductive switching, thereby protecting the circuit components. This straightforward microcontroller-based design supports reliable operation and is suitable for embedded and near real-time assistive applications.

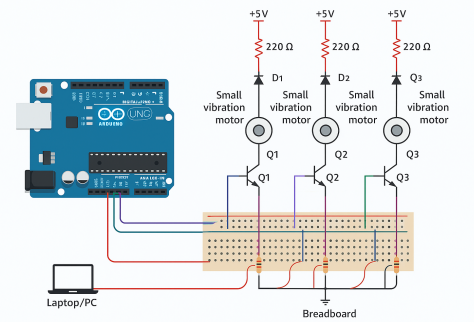


Fig. 2. Circuit diagram of the electromechanical vibration motor interface controlled by Arduino Uno

C. Software–Hardware Integration

The multimodal inference module and the Arduino microcontroller are in constant interaction with one another using USB serial communication. Once the fusion model classifies semantically the input, it sends a control signal corresponding to the category to the Arduino. Then the Arduino unit activates the specific vibration pattern related to the predicted class based on the received signal.

The integration of software and hardware in the system leads to negligible delay between the processes of classification and tactile feedback generation that consequently supports near real-time operation. The modular structure of the system not only provides a way for easy alteration in the future but also permits the addition of new input modalities without extensive changes to the hardware layout.

V. RESULTS AND DISCUSSION

The study proposes a multimodal tactile communication system which now undergoes testing in this section. The analysis evaluates multiple performance aspects which include classification accuracy based on the derived semantic labels and testing results of multimodal fusion and baseline model performance and testing results of tactile feedback interpretation system.

A. Tactile Interpretation Evaluation

The team conducted a user study to test how well people could understand the vibration patterns that they created. The participants received a brief demonstration which showed them how three semantic categories—Instruction, Alert, and Information—mapped to their respective vibration patterns before the testing process started. The testing procedure required users to identify the semantic category which matched the vibration outputs that the system produced.

A prediction was considered correct if the participant’s response matched the semantic class produced by the multimodal classifier. The study found tactile interpretation accuracy because test subjects correctly identified vibration patterns during all testing situations. The evaluation examines how well users can understand and use tactile mappings but does not measure learning success when compared to the actual data from the dataset.

B. Model Performance Evaluation

The evaluation of the multimodal fusion model was done on a test dataset that was held out from the beginning and having standard classification metrics such as accuracy, precision, recall, and F1-score applied for three semantic categories namely: Instruction, Alert, and Information.

The proposed fusion model’s accuracy on the test dataset was **91%**, thus confirming the robustness of the model across the different kinds of image–text pairs. The performance was also uniform throughout the semantic classes suggesting that the model does not have any bias towards a certain category. This improvement reflects the complementary contribution of visual and textual features.

C. Confusion Matrix Analysis

The confusion matrix for the multimodal fusion model is presented in Figure 3. The strong diagonal dominance reflects good classification across all semantic categories.

True Label/Predicted Label	Instruction	Alert	Info
Instruction	3377	6	109
Alert	2	48	32
Info	76	23	4418

Fig. 3. Confusion matrix of the fine-tuned multimodal fusion model

The Instruction and Alert classes showed very few misclassifications, and these were mainly due to the semantic overlap in their short textual descriptions. The Information class had the largest classification accuracy, as it was supported with more noticeable contextual patterns.

D. Training and Validation Analysis

Figures 4 and 5 show the training loss and validation accuracy curves of the fusion model, respectively. The training loss was gradually reduced, and the validation accuracy remained stable, which is a sign of effective learning and that overfitting did not occur.

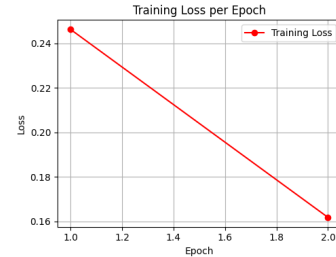


Fig. 4. Training loss curve of the fusion model

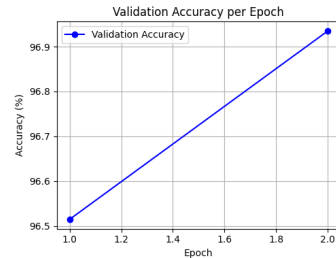


Fig. 5. Validation accuracy curve of the fusion model

E. Comparison with Baseline Models

The importance of deploying multimodal fusion was highlighted with the proposed system being compared to unimodal baseline models that depended solely on text or

TABLE V
COMPARISON OF MULTIMODAL FUSION MODEL WITH BASELINE
APPROACHES

Model	Accuracy	F1-Score
BERT (Text-only)	78%	75%
LSTM (Text-only)	65%	62%
Lightweight CNN (Image-only)	68%	64%
Fusion (CNN + BERT + LSTM)	91%	90%

image inputs, respectively. A detailed comparison of results is shown in Table V.

The fusion-based method was superior to all unimodal baselines, and it was able to provide a **13%** increment over the text-only model and a **23%** increment over the image-only model. The results clearly show that the combination of visual and textual modalities leads to more powerful semantic understanding by using the strengths of both sources.

F. Tactile Feedback Interpretation

An evaluation at the user level was conducted to measure how comprehensible the produced patterns of vibrations were. After a short familiarization period the participants were asked to point out the semantic category that was transmitted through the tactile feedback. The average accuracy of the tactile interpretation was **92%**, which means that the vibration patterns were intuitively understood and easily differentiated.

Moreover, the time lag between semantic classification and tactile activation was so small that it could be observed as minimal. This supports the applicability of the proposed system in time-sensitive assistive scenarios.

VI. CONCLUSION AND FUTURE WORK

The research paper presented a tactile feedback system which implements vibration technology to deliver information through multimodal learning that combines visual and textual information. The system implemented a lightweight convolutional neural network for visual data processing while it used BERT embeddings and a BiLSTM network to extract textual data. The system used feature-level fusion to link visual and textual data which produced semantic classification results that corresponded with specific vibration patterns of an electromechanical tactile device.

The study method produced accurate semantic classification results which allowed users to understand vibration patterns after they completed brief training. The system displays its operational capabilities through its combination of lightweight models and inexpensive hardware which allows multimodal tactile communication to operate in assistive environments that require near real-time performance without visual or auditory communication.

The upcoming research will work on developing additional semantic labels which will enable the creation of complex tactile communication systems that use various contextual elements. The study will include user tests which will involve participants from different backgrounds to measure their ability to use the system over time and how they

acquire new knowledge. The system will receive dual enhancements through two main improvements which include the development of adaptive vibration patterns and the creation of embedded systems while new input channels will be added to boost system stability and performance across different environments.

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